CAFÉ SALES REPORT





INTRODUCTION

This analysis explores sales performance from the Café Sales dataset.
Data was cleaned and visualized in Python to uncover trends in:

- Quantity purchased
- Revenue generated
- Payment methods
- Product performance

DATA CLEANING



01 Filled missing numeric values with the mean.

02 Replaced invalid entries like "ERROR", "UNKNOWN" and NaN with appropriate values (‘Unspecified’).

03 Ensured all dates were in a consistent format.

04 Feature Engineering: Created new columns, such as Day of the Week or Transaction Month, for further analysis.

05 Standardized and crosschecked data types for all columns

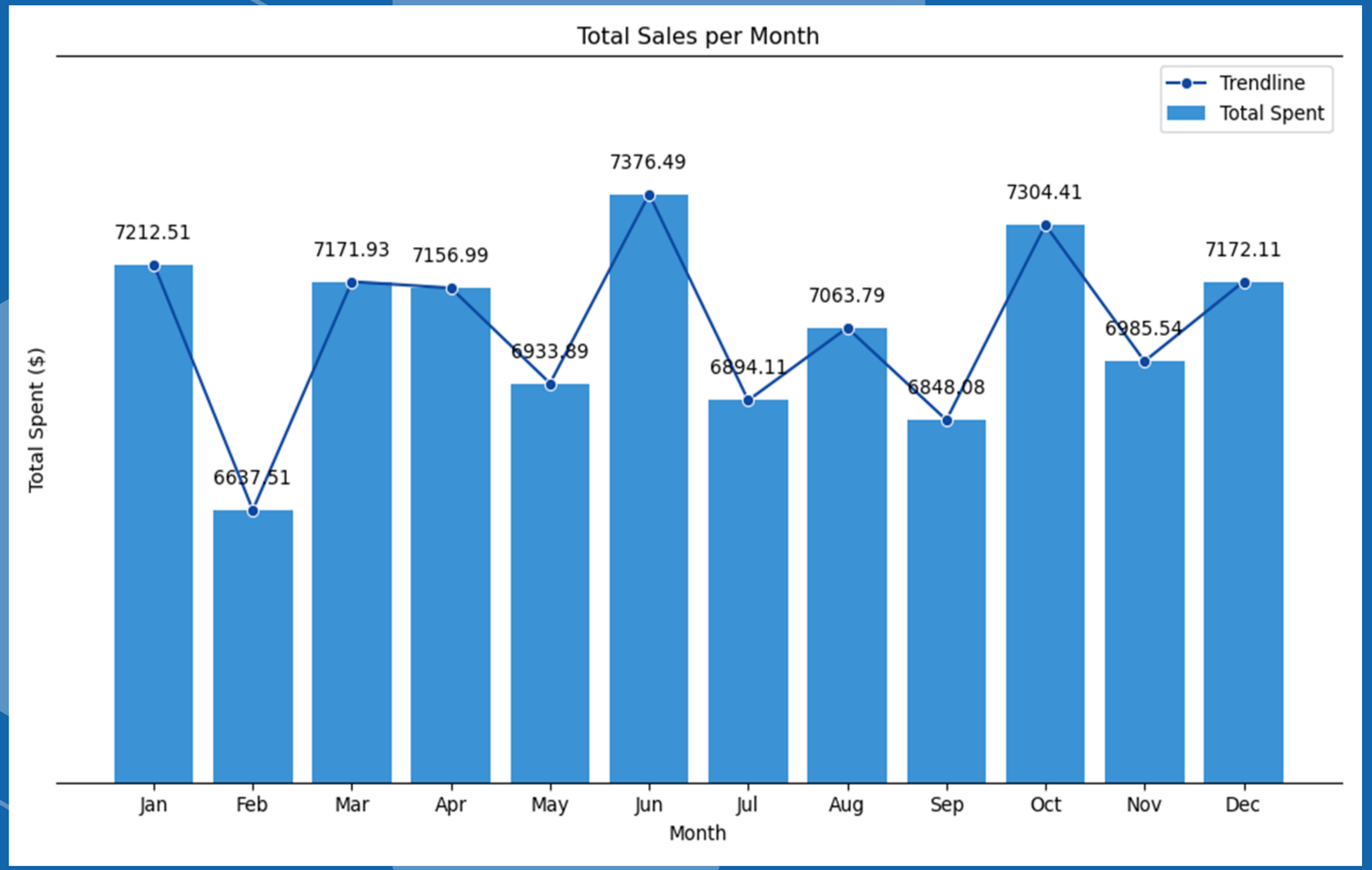
OBJECTIVES OF THE PROJECT



1. Analyze total sales over time (daily, weekly, or monthly).
2. Identify the best-selling items by quantity and total revenue.
3. Compare total sales across different product categories (e.g., Coffee, Pastry, Drinks).
4. Determine the most popular payment methods (e.g., Cash, Card, Mobile).
5. Identify the factors that significantly affect total revenue.

DESCRIPTIVE AND DIAGNOSTIC ANALYTICS:

Objective 1: Total Sales Over Time

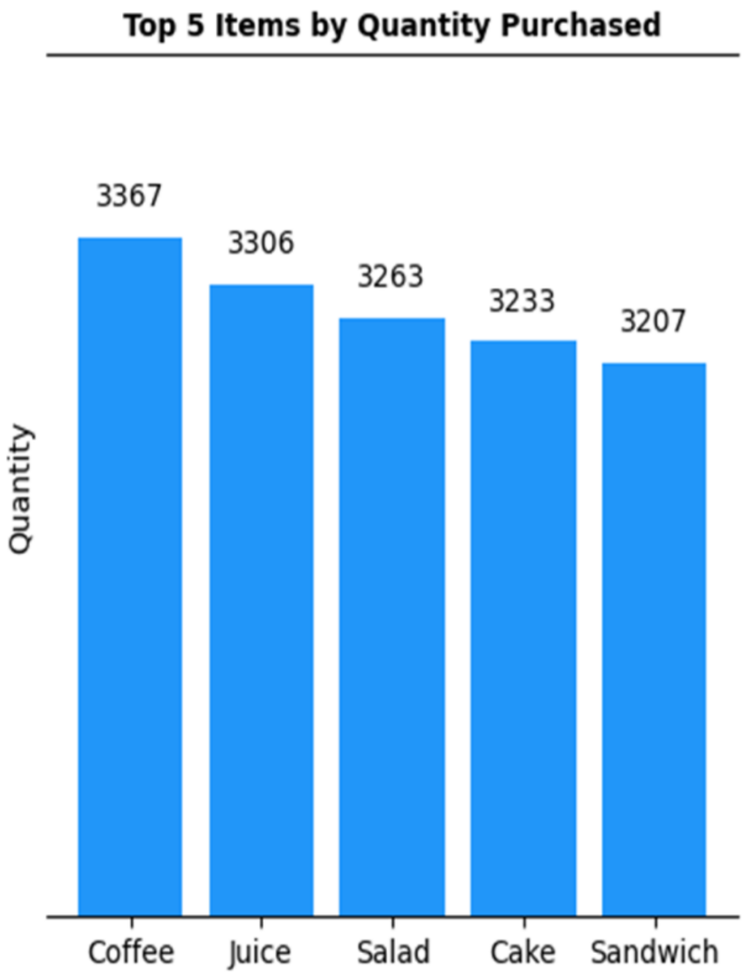


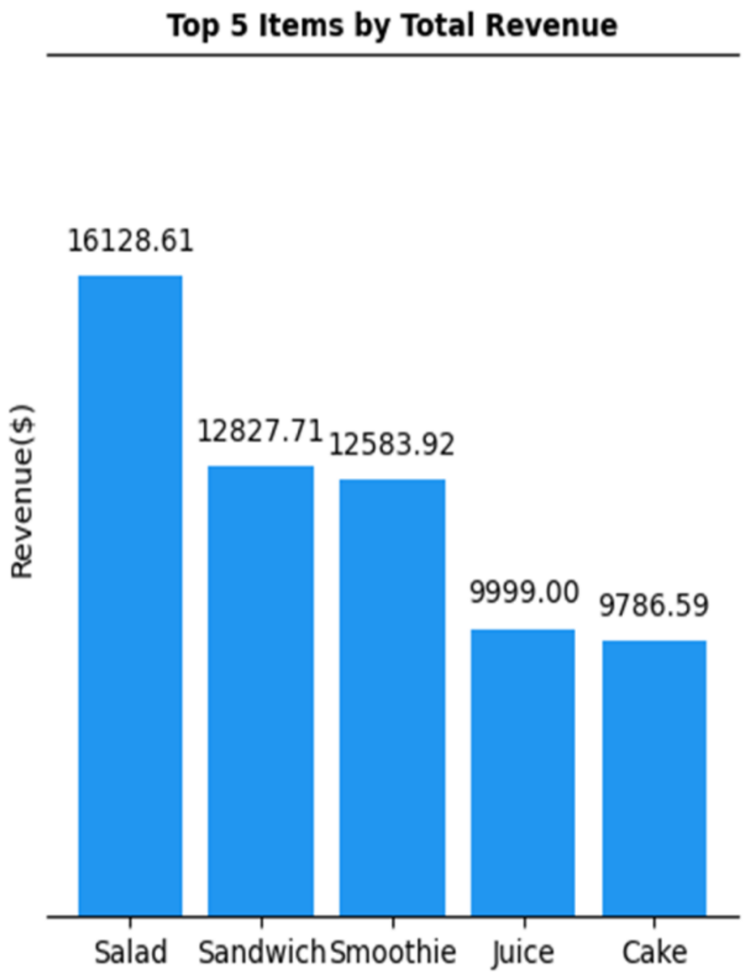
INSIGHTS:

Sales was highest in the month of June.

There was a similar spike in sales from September to October.

Objective 2: Best-selling Items by Quantity and Total Revenue.

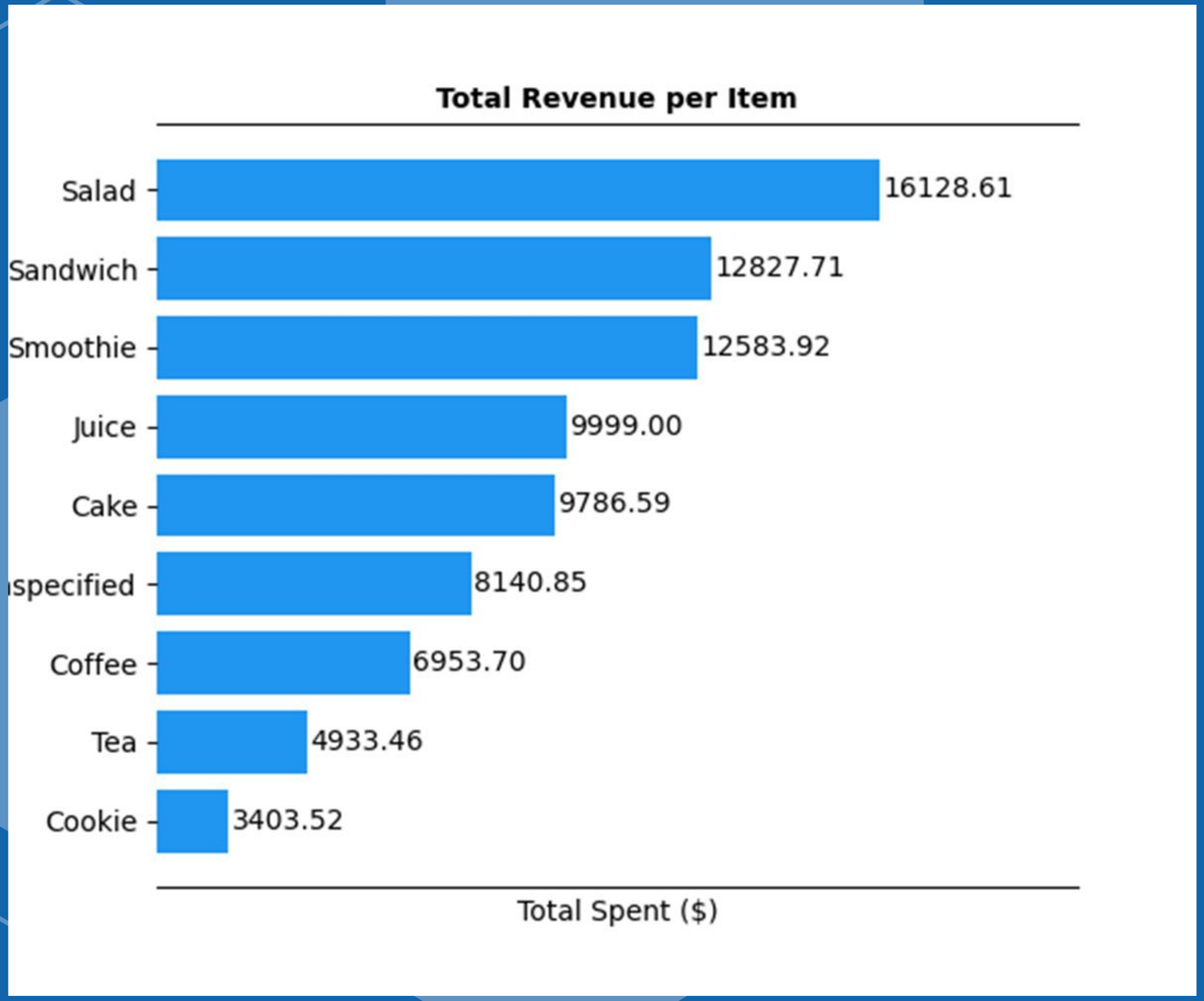




INSIGHTS:

Coffee and Salad were the best-selling items per Quantity and Total Revenue respectively.

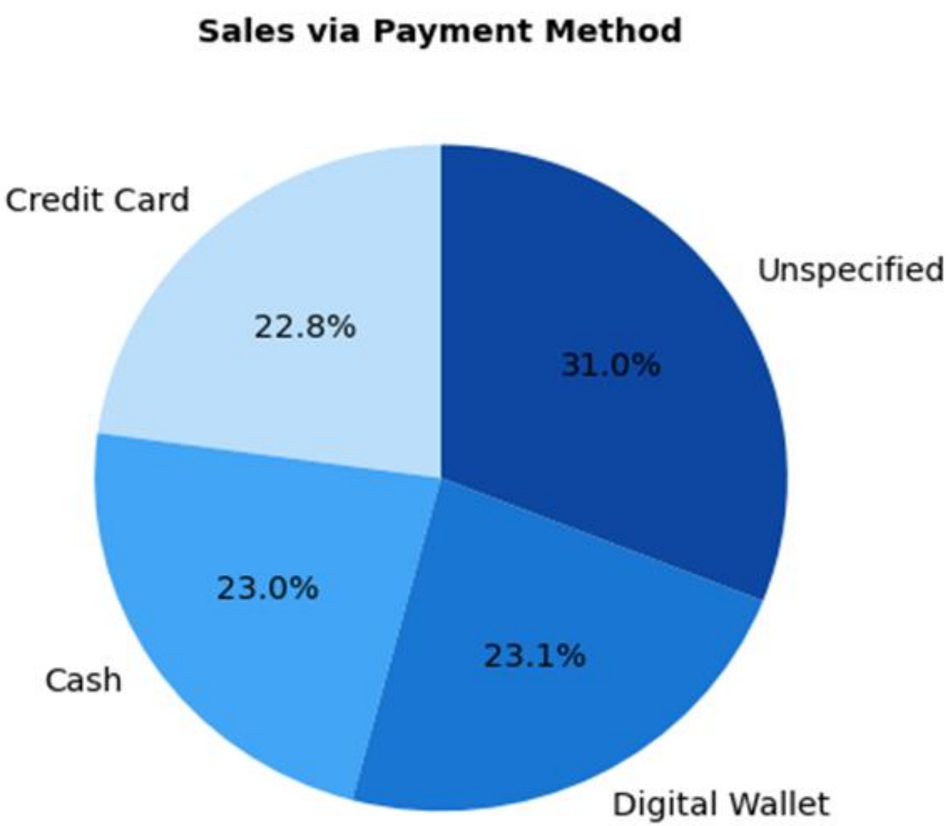
Objective 3: Comparing Total Sales across Different Product Categories



INSIGHTS:

Salad generated the highest revenue ($16,128.61), followed by Sandwich and Smoothie; meaning that these products drive the most income for the business.

Objective 4: Most Popular Payment Methods

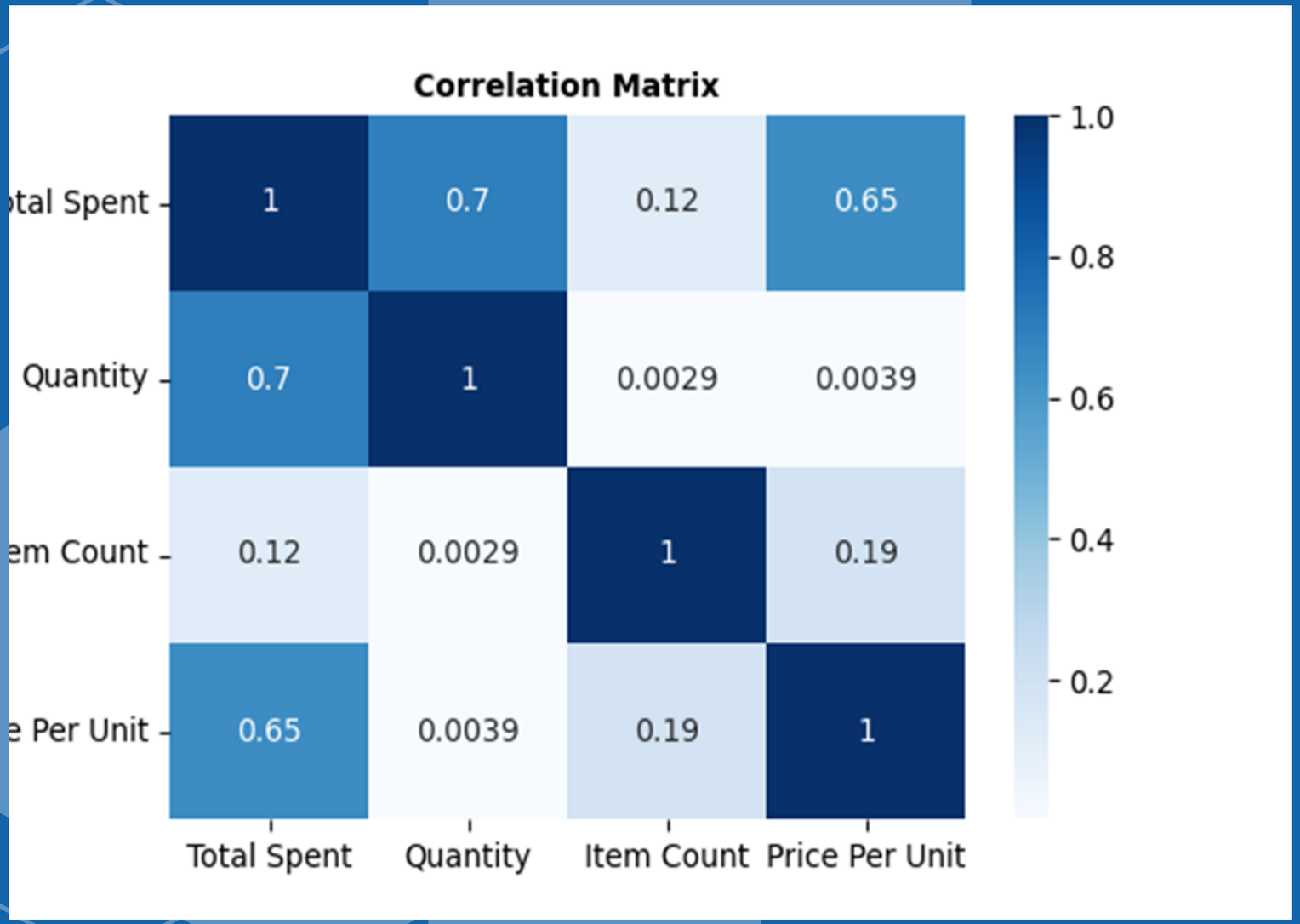


INSIGHTS:

Sales payments were made through four channels as indicated in the pie chart, with usage patterns nearly equal and Digital Wallet barely leading.

There was however, a large portion of transactions that did not specify the method, accounting for 31% of the data.

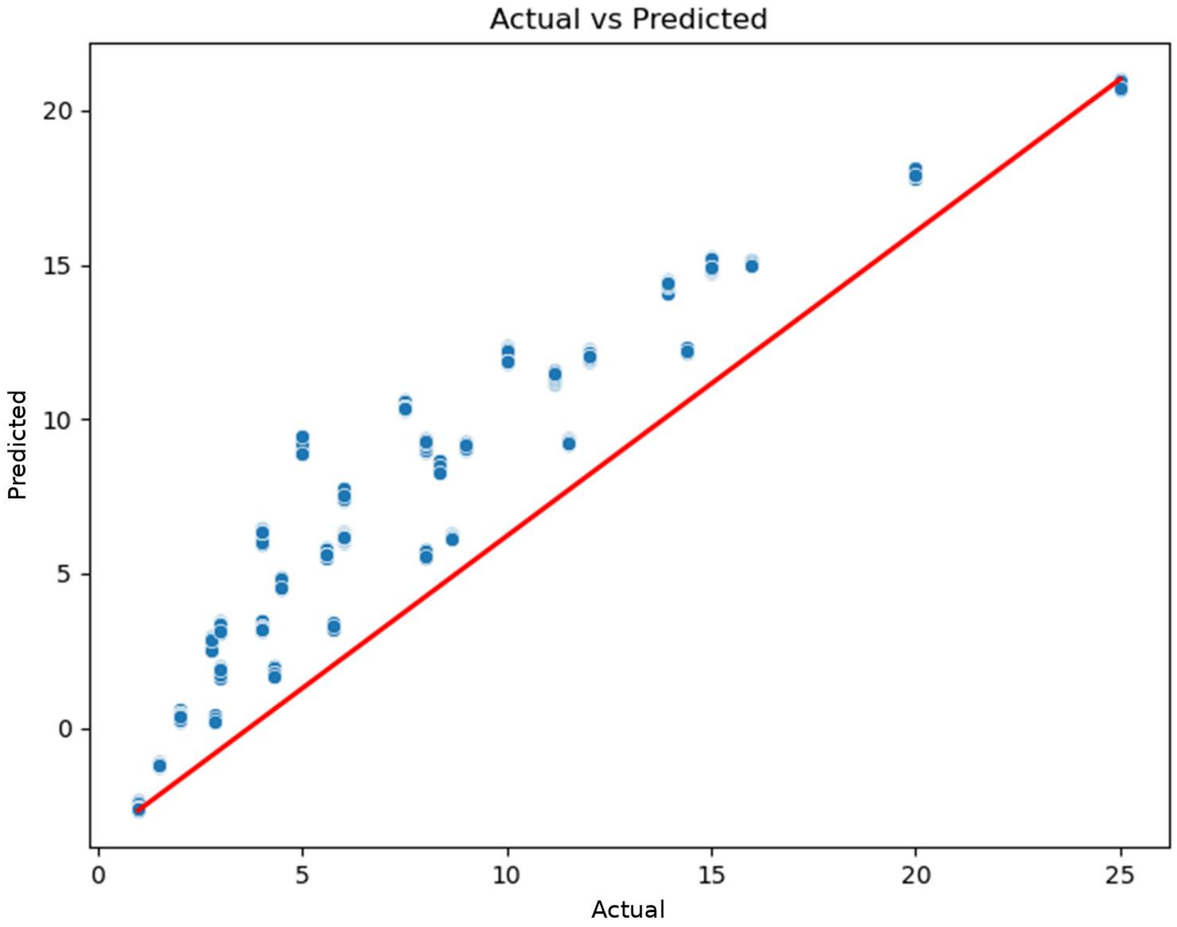
Objective 5: Factors That Significantly Affect Total Revenue.



INSIGHTS:

The correlation matrix shows a strong positive relationship (0.7) between Total Spent and Quantity, and between Total Spent and Price Per Unit (0.65). This suggests that both higher prices and higher purchase volumes contribute significantly to total revenue.

PREDICTIVE ANALYTICS

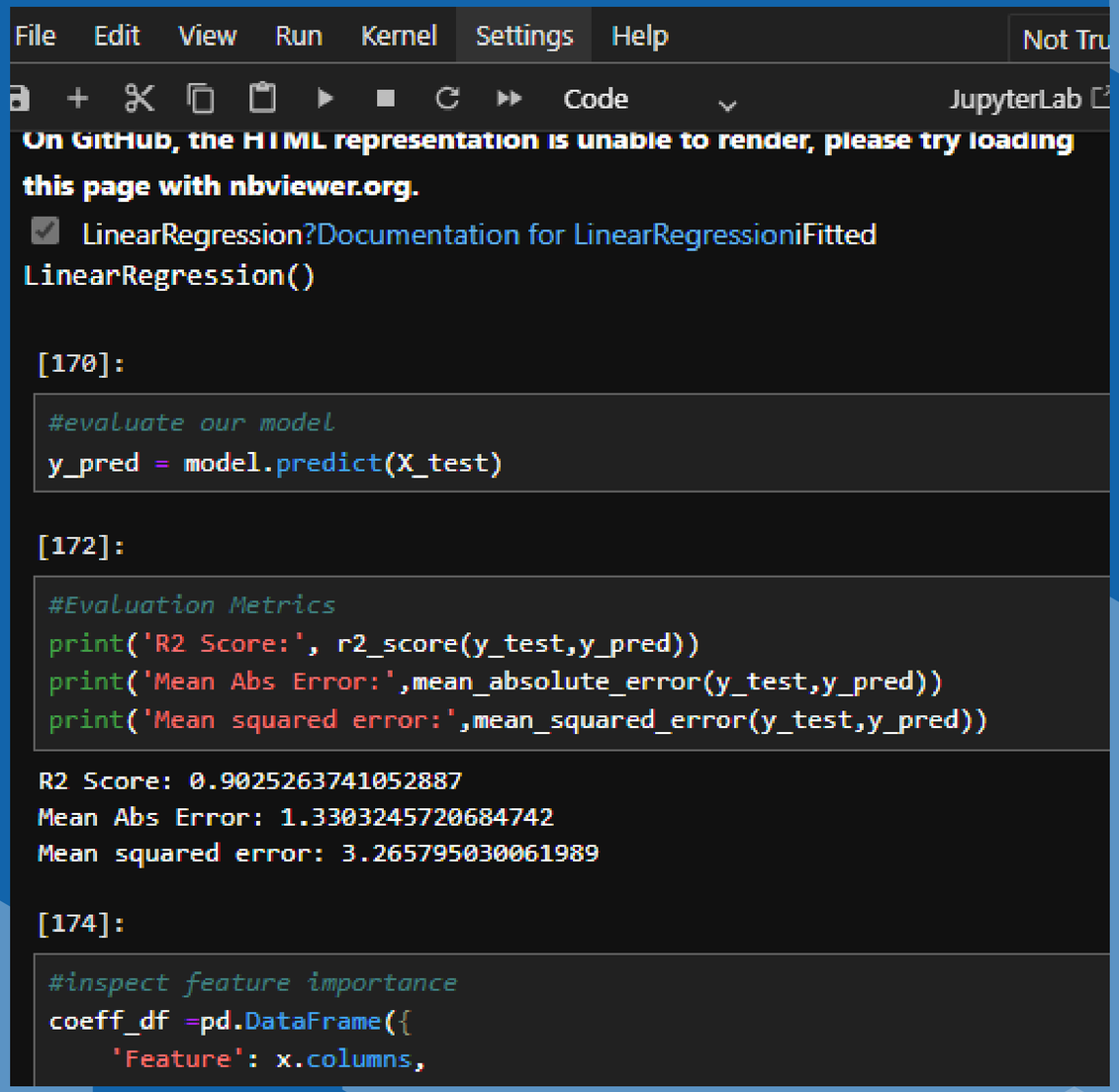


Goal: predict how much a customer will spend.

Method: Linear Regression, using Item, Payment Method and Location as predictors.

Predicted values closely follow actual values across the range, consistent with the model's R^2 of 0.90.

PREDICTIVE ANALYTICS



We obtained Evaluation Metrics of:

R2 Score: 0.9025263741052887

Mean Abs Error: 1.3303245720684742

Mean squared error: 3.265795030061989

These metrics confirm that the forecasts from the model are reliable.

PRESCRIPTIVE ANALYSIS

Recommended Business Actions

01 Promote high-revenue items like Salad, Sandwich and Smoothie to increase revenue.

02 Bundle best-selling with low-revenue items (e.g, Coffee and Cookie) to increase sales.

03 Introduce loyalty discounts for high-frequency buyers.

04 Optimize pricing: slightly reduce prices on low-selling items to attract more customers.

05 Improve stock management by ensuring consistent availability of top-performing products.

CONCLUSION

This analysis shows that Salad, Sandwich and Smoothie are the café's main revenue drivers, with demand peaking from September to October. The predictive model estimates customer spend reliably, and if current patterns continue, sales should remain steady. By promoting top earners, bundling popular items and keeping key products in stock, the café can turn these insights into better decisions.



THANK YOU

